Beginning with the general solution derive the particular solution:

$$u(x,t) = \frac{1}{2}[\psi(x+t) + \psi(x-t)] + \frac{1}{2}\int_{x-t}^{x+t} \psi(\tau)d\tau$$

HINT: At one point you will want to integrate $F'(x)$. It helps if you use the notation: $\int_a^x F'(\tau)d\tau = F(x) - F(a)$.

(b) (6 pts) Show by direct substitution that the particular solution satisfies the initial value problem.

8. (8 pts) Boundary Condition Matching problem (see last 2 pages). Feel free to print out the problem and then do your work on those sheets of papers so that you do not need to reproduce the graphs.

Matching problem:

Consider the 1-dimensional heat equation with the following four different boundary conditions:

D (i) u(0,t)=1 u(1,t)=2

E (ii) u(0,t)=1 (partial u)/(partial x) (1,t) = 0

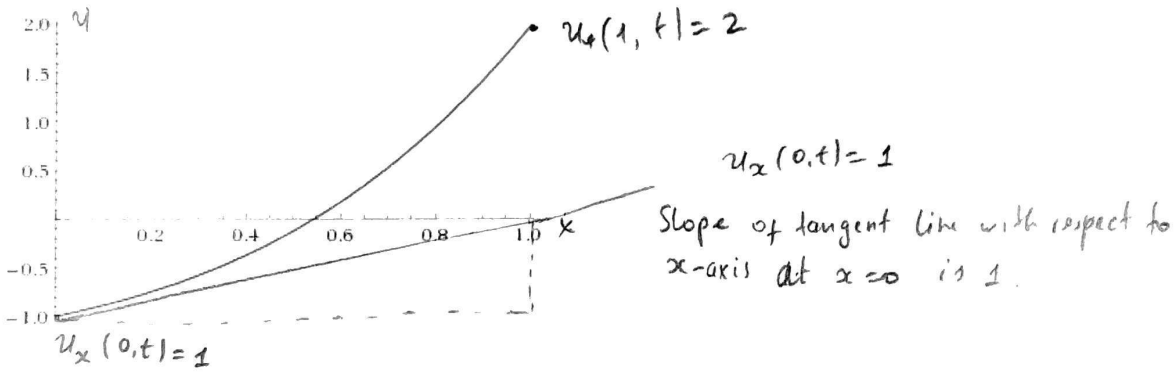
A (iii) (partial u)/(partial x) (0,t) = 1 u(1,t) = 2

B (iv) (partial u)/(partial x) (0,t) = 0 (partial u)/(partial x)(1,t)=0

You asked a friend to plot the solutions to the heat equation with the above boundary conditions, but your friend forgot to label the graphs! Below are the graphs generated at some time t (not necessarily at equilibrium). Use your knowledge about how the solution should behave at the boundaries to match the above boundary conditions with the following graphs. Note that there is one more graph than set of boundary conditions (so your friend must have generated an extra plot from something).

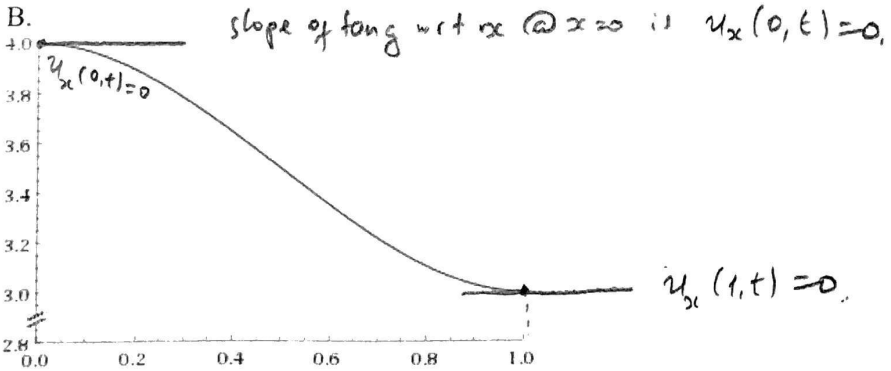
All graphs are of temperature at some time t. The horizontal axis is x, and the vertical axis is temperature. (Plotted using WolframAlpha).

A.

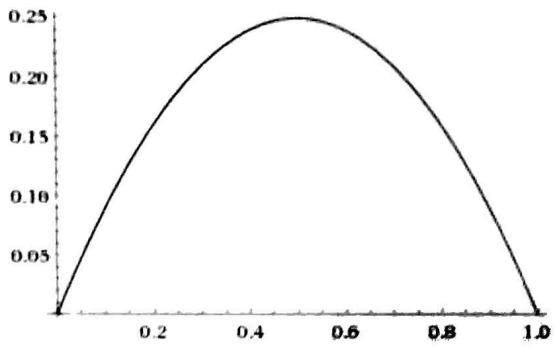


B.

(iv)

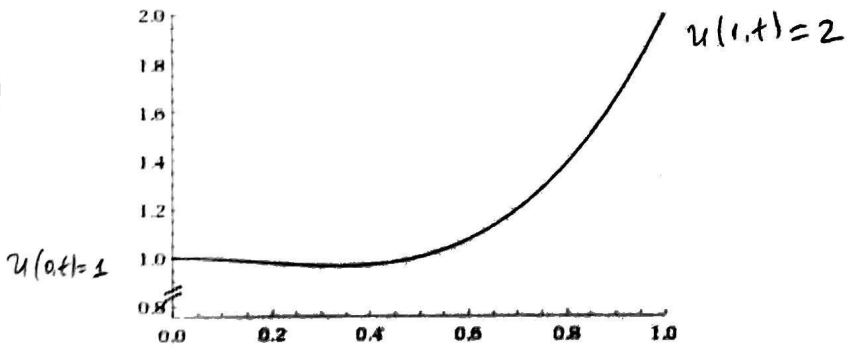


C.


0.25
0.20
0.15
0.10
0.05
0.2
0.4
0.6
0.8
1.0

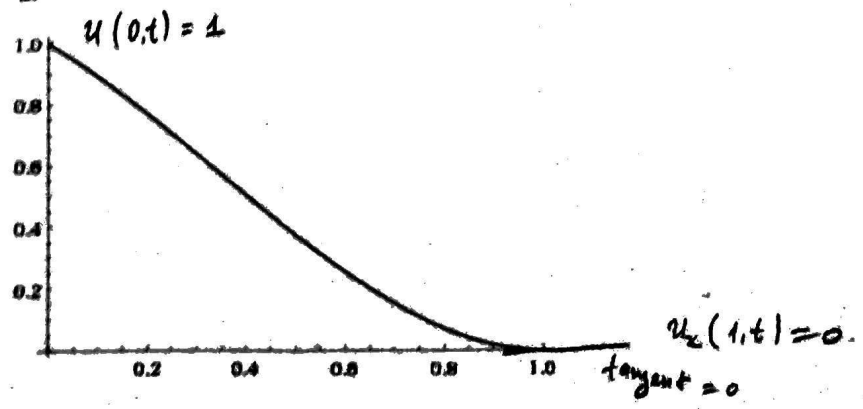
D.

(i)


2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.0
0.2
0.4
0.6
0.8
1.0
u(1,t)=2
u(0,t)=1

E.

(ii)


u(0,t) = 1
1.0
0.8
0.6
0.4
0.2
0.2
0.4
0.6
0.8
1.0
u_x(1,t)=0.
tangent = 0